



EXAM PAPERS PRACTICE

Boost your performance and confidence with these topic-based exam questions

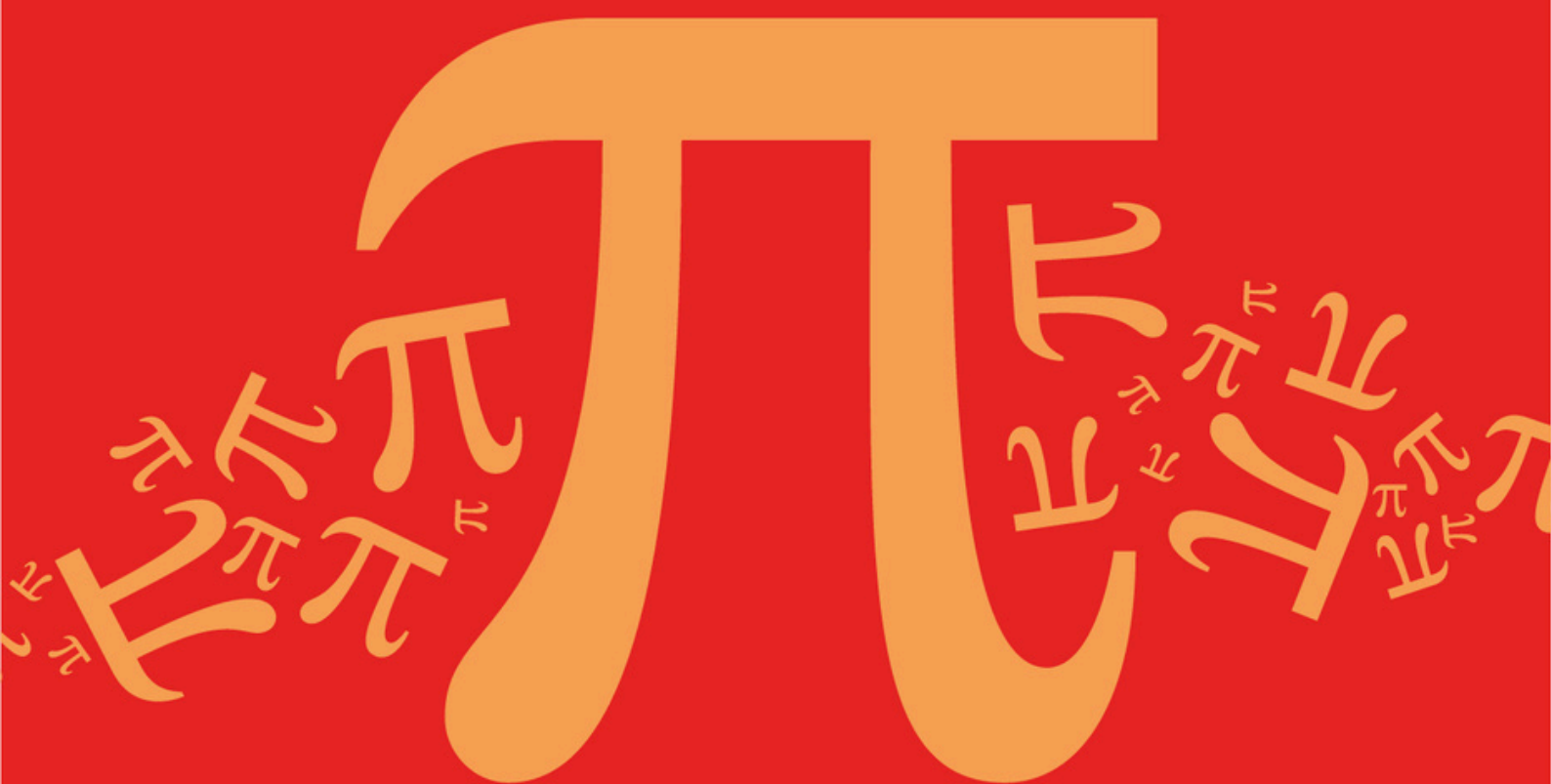
Practice questions created by actual examiners and assessment experts

Detailed mark scheme

Suitable for all boards

Designed to test your ability and thoroughly prepare you

AQA A Level Further Mathematics (7367)



Topic

C: Matrices

Sub topic

C5: Determinants (3x3 Determinants)

Topic Questions

AQA A Level Further Mathematics (7367)

Question Paper

Topic

C: Matrices

Sub topic

C5: Determinants (3x3 Determinants)

EXAM PAPERS PRACTICE

© 2024 Exams Papers Practice. All Rights Reserved

**To be used by all students and teachers for
AQA A Level Further Mathematics (7367)
for exams from 2025 onwards.**

Students and teachers of other boards may also find this useful

Name : _____

Date : _____

Time : _____

Total Marks Available : _____

Total Marks Received : _____

Q1.

Three planes have equations

$$x - y + kz = 3$$

$$kx - 3y + 5z = -1$$

$$x - 2y + 3z = -4$$

where k is a real constant. The planes do not meet at a unique point.

- (a) Find the possible values of k

(3)

- (b) There are two possible geometric configurations for the given planes.

Identify each possible configuration, stating the corresponding value of k

Fully justify your answer.

(5)

- (c) Given further that the equations of the planes form a consistent system, find the solution of the system of equations.

(3)

(Total 11 marks)

Q2.

It is given that **A** and **B** are 3×3 matrices such that

$$\det(\mathbf{AB}) = 24 \text{ and } \det(\mathbf{A}^{-1}) = -3$$

- (a) State the value of $\det \mathbf{A}$.

(1)

- (b) A three-dimensional shape S , with volume 20 cm^3 , is transformed using matrix **B**.

Find the volume of the image of S .

(3)

(Total 4 marks)

Q3.

The system of equations

$$2x - y - z = 3$$

$$x + 2y - 3z = 4$$

$$2x + y + az = b$$

does not have a unique solution.

- (a) Show that $a = -3$.

(3)

- (b) Given further that the equations are inconsistent, find the possible values of b .

(2)

(Total 5 marks)

Q4.

The determinant Δ is given by

$$\Delta = \begin{vmatrix} x^2 - x & y^2 - y & z^2 - z \\ x & y & z \\ x^2 + y^2 + z^2 & x^2 + y^2 + z^2 & x^2 + y^2 + z^2 \end{vmatrix}$$

where x , y and z are distinct real numbers.

- (a) Express Δ as a product of one quadratic factor and three linear factors.

(6)

- (b) Deduce that $\Delta \neq 0$.

(2)

(Total 8 marks)

Q5.

The matrix $\mathbf{M}$ is given by

$$\mathbf{M} = \begin{bmatrix} 1 & 1 & 2 \\ 0 & 2 & 2 \\ -1 & 1 & 3 \end{bmatrix}$$

- (a) Show that $\lambda = 2$ is an eigenvalue for $\mathbf{M}$, and find the other two eigenvalues.

(5)

- (b) Find an eigenvector that corresponds to $\lambda = 2$.

(3)

- (c) The matrix $\mathbf{N}$ is given by

$$\mathbf{N} = \begin{bmatrix} 2 & 2 & -3 \\ 2 & 2 & 3 \\ -3 & 3 & 3 \end{bmatrix}$$

- (i) Show that $\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ is an eigenvector for $\mathbf{N}$, and find the corresponding eigenvalue.

(2)

- (ii) Hence state one eigenvector for the matrix **MN**, and find the corresponding eigenvalue.

(3)

(Total 13 marks)

Q6.

The 3×3 matrices **A** and **B** satisfy

$$\mathbf{AB} = \begin{bmatrix} k & 8 & 1 \\ 1 & 1 & 0 \\ 1 & 4 & 0 \end{bmatrix}, \text{ where } \mathbf{A} = \begin{bmatrix} k & 6 & 8 \\ 0 & 1 & 2 \\ -3 & 4 & 8 \end{bmatrix}$$

and k is a constant.

- (a) Show that **AB** is non-singular.

(1)

- (b) Find $(\mathbf{AB})^{-1}$ in terms of k .

(5)

- (c) Find $\mathbf{B}^{-1}$.

(4)

(Total 10 marks)

Q7.

- (a) Determine the two values of the integer n for which the system of equations

$$\begin{aligned} 2x + ny + z &= 5 \\ 3x - y + nz &= 1 \\ -x + 7y + z &= n \end{aligned}$$

does not have a unique solution.

(4)

- (b) For the positive value of n found in part (a), determine whether the system is consistent or inconsistent, and interpret this result geometrically.

(6)

(Total 10 marks)

Q8.

$$\text{Let } \Delta = \begin{vmatrix} yz & xz & xy \\ x & y & z \\ x^2 & -y^2 & z^2 \end{vmatrix}.$$

- (a) Show that $(y + z)$ is a factor of Δ .

(2)

- (b) Factorise Δ as completely as possible.

(4)

(Total 6 marks)

Q9.

The matrix $\mathbf{A} = \begin{bmatrix} k & 1 & 2 \\ 2 & k & 1 \\ 1 & 2 & k \end{bmatrix}$, where k is a real constant.

- (a) (i) Show that there is a value of k for which

$$\mathbf{A} \mathbf{A}^T = m \mathbf{I}$$

where m is a rational number to be determined and $\mathbf{I}$ is the 3×3 identity matrix.

(6)

- (ii) Deduce the inverse matrix, $\mathbf{A}^{-1}$, of $\mathbf{A}$ for this value of k .

(1)

- (b) (i) Find $\det \mathbf{A}$ in terms of k .

(2)

- (ii) In the case when $\mathbf{A}$ is singular, find the integer value of k and show that there are no other possible real values of k .

(3)

- (iii) Find the value of k for which $\lambda = 7$ is a real eigenvalue of $\mathbf{A}$.

(2)

(Total 14 marks)

Q10.

Let $\Delta = \begin{vmatrix} 1 & 2 & 3 \\ x & y & z \\ y+z & z+x & x+y \end{vmatrix}$.

- (a) Use a row operation to show that $(x + y + z)$ is a factor of Δ .

(2)

- (b) Hence, or otherwise, express Δ as a product of linear factors.

(2)

(Total 4 marks)

Q11.

The matrices $\mathbf{A}$ and $\mathbf{B}$ are given in terms of p by

$$\mathbf{A} = \begin{bmatrix} 1 & p & 4 \\ -3 & 2 & 1 \\ 2 & -1 & 1 \end{bmatrix} \quad \text{and} \quad \mathbf{B} = \begin{bmatrix} p & 1 & 5 \\ 9 & p & -1 \\ 2 & 0 & 1 \end{bmatrix}$$

- (a) Find each of $\det \mathbf{A}$ and $\det \mathbf{B}$ in terms of p .

(3)

- (b) Without finding $\mathbf{AB}$, determine all values of p for which $\mathbf{AB}$ is singular.

(3)

(Total 6 marks)

Q12.

$$\Delta = \begin{vmatrix} n(n+1) & n+1 & -1 \\ 0 & 1 & n \\ 1 & -(n+1) & 1 \end{vmatrix}$$

Let

- (a) (i) Show that $(n^2 + n + 1)$ is a factor of Δ .

(2)

- (ii) Hence, or otherwise, express Δ in factorised form.

(2)

- (b) By expanding Δ directly, show that

$$\Delta = [n(n+1)]^2 + f(n)$$

where $f(n)$ can be expressed as the sum of two squares.

(2)

- (c) Hence express the number 12 321 as the sum of three squares.

(2)

(Total 8 marks)

Q13.

$$\Delta = \begin{vmatrix} 16-q & 5 & 7 \\ -12 & -1-q & -7 \\ 6 & 6 & 10-q \end{vmatrix}$$

- (a) It is given that

- (i) By using row operations on the first two rows of Δ , show that $(4 - q)$ is a factor of Δ .

(2)

- (ii) Express Δ as the product of three linear factors.

(4)

(b) It is given that $\mathbf{M} = \begin{bmatrix} 16 & 5 & 7 \\ -12 & -1 & -7 \\ 6 & 6 & 10 \end{bmatrix}$.

- (i) Verify that $\begin{bmatrix} 2 \\ 5 \\ -7 \end{bmatrix}$ is an eigenvector of $\mathbf{M}$ and state its corresponding eigenvalue. (3)
- (ii) For each of the other two eigenvalues of $\mathbf{M}$, find a corresponding eigenvector. (7)
- (c) The transformation T has matrix $\mathbf{M}$. Write down cartesian equations for any one of the invariant lines of T . (2)
- (Total 18 marks)

Q14.

- Factorise fully the determinant $\begin{vmatrix} x & y & z \\ x^2 & y^2 & z^2 \\ yz & zx & xy \end{vmatrix}$. (Total 8 marks)

Q15.

- (a) Expand the determinant

$$\mathbf{D} = \begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ z & x & y \end{vmatrix}$$

- (b) Show that $(x + y + z)$ is a factor of the determinant

$$\Delta = \begin{vmatrix} x & y & z \\ y-z & z-x & x-y \\ x+z & y+x & z+y \end{vmatrix}$$

- (c) Show that $\Delta = k(x + y + z)D$ for some integer k .

(3)
(Total 7 marks)

Q16.

- (a) Matrix $\mathbf{M} = \begin{bmatrix} a & b & c \\ c & a & b \\ a & c & a \end{bmatrix}$. Without attempting to factorise, expand fully $\det \mathbf{M}$.

(2)

- (b) Matrix $\mathbf{N} = \begin{bmatrix} d & e & f \\ f & d & e \\ e & f & d \end{bmatrix}$. Find the product matrix $\mathbf{MN}$. (3)

- (c) Prove that the product

$$(a^3 + b^3 + c^3 - 3abc)(d^3 + e^3 + f^3 - 3def)$$

can be written in the form $x^3 + y^3 + z^3 - 3xyz$, stating clearly each of x , y and z in terms of a , b , c , d , e and f .

(2)
(Total 7 marks)

Q17.

A system of equations is given by

$$\begin{aligned} x + 3y + 5z &= -2 \\ 3x - 4y + 2z &= 7 \\ ax + 11y + 13z &= b \end{aligned}$$

where a and b are constants.

- (a) Find the unique solution of the system in the case when $a = 3$ and $b = 2$. (5)
- (b) (i) Determine the value of a for which the system does not have a unique solution. (3)
- (ii) For this value of a , find the value of b such that the system of equations is consistent. (4)

(Total 12 marks)

Q18.

The matrix $\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 3 \\ 4 & 3 & k \end{bmatrix}$, where k is a constant.

Determine, in terms of k where appropriate:

- (a) $\det \mathbf{A}$; (2)
- (b) $\mathbf{A}^{-1}$. (5)

(Total 7 marks)

Q19.

A transformation T of three-dimensional space is given by the matrix $\mathbf{W} = \begin{bmatrix} 3 & -1 & 1 \\ 2 & 0 & 2 \\ -1 & 1 & 1 \end{bmatrix}$.

- (a) (i) Evaluate $\det \mathbf{W}$, and describe the geometrical significance of the answer in relation to T .

(2)

- (ii) Determine the eigenvalues of $\mathbf{W}$.

(6)

- (b) The plane H has equation $\mathbf{r} \cdot \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} = 0$.

- (i) Write down a cartesian equation for H .

(1)

- (ii) The point P has coordinates (a, b, c) . Show that, whatever the values of a , b and c , the image of P under T lies in H .

(4)

(Total 13 marks)

Q20.

By considering the determinant

$$\begin{vmatrix} x & y & z \\ z & x & y \\ y & z & x \end{vmatrix}$$

show that $(x + y + z)$ is a factor of $x^3 + y^3 + z^3 - kxyz$ for some value of the constant k to be determined.

(Total 3 marks)

Q21.

- (a) Show that $(a - b)$ is a factor of the determinant

$$\Delta = \begin{vmatrix} a & b & c \\ b+c & c+a & a+b \\ bc & ca & ab \end{vmatrix}$$

(2)

- (b) Factorise Δ completely into linear factors.

(5)

(Total 7 marks)

Q22.

The matrix $\mathbf{P} = \begin{bmatrix} 4 & -1 & 2 \\ 1 & 1 & 3 \\ -2 & 0 & a \end{bmatrix}$, where a is constant.

- (a) (i) Determine $\det \mathbf{P}$ as a linear expression in a . (2)
- (ii) Evaluate $\det \mathbf{P}$ in the case when $a = 3$. (1)
- (iii) Find the value of a for which $\mathbf{P}$ is singular. (2)
- (b) The 3×3 matrix $\mathbf{Q}$ is such that $\mathbf{PQ} = 25\mathbf{I}$.
Without finding $\mathbf{Q}$:
- (i) write down an expression for $\mathbf{P}^{-1}$ in terms of $\mathbf{Q}$; (1)
- (ii) find the value of the constant k such that $(\mathbf{PQ})^{-1} = k\mathbf{I}$; (2)
- (iii) determine the numerical value of $\det \mathbf{Q}$ in the case when $a = 3$. (4)

(Total 12 marks)

Q23.

©2025 Exam Papers Practice. All rights reserved.

Factorise completely the determinant $\begin{vmatrix} y & x & x+y-1 \\ x & y & 1 \\ y+1 & x+1 & 2 \end{vmatrix}$.

(Total 6 marks)

Q24.

The matrices $\mathbf{P}$ and $\mathbf{Q}$ are defined in terms of the constant k by

$$\mathbf{P} = \begin{bmatrix} 3 & 2 & 1 \\ 1 & -1 & k \\ 5 & 3 & 2 \end{bmatrix} \text{ and } \mathbf{Q} = \begin{bmatrix} 5 & 4 & 1 \\ 3 & k & -1 \\ 7 & 3 & 2 \end{bmatrix}$$

- (a) Express $\det \mathbf{P}$ and $\det \mathbf{Q}$ in terms of k . (3)
- (b) Given that $\det(\mathbf{PQ}) = 16$, find the two possible values of k . (4)

(Total 7 marks)

Q25.

A transformation is represented by the matrix $\mathbf{A} = \begin{bmatrix} 0.28 & -0.96 & 0 \\ 0.96 & 0.28 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

- (a) Evaluate $\det \mathbf{A}$. (1)
 - (b) State the invariant line of the transformation. (1)
 - (c) Give a full geometrical description of this transformation. (3)
- (Total 5 marks)

Q26.

Express the determinant $\begin{vmatrix} 1 & a^2 & bc \\ 1 & b^2 & ca \\ 1 & c^2 & ab \end{vmatrix}$ as the product of four linear factors.

(Total 6 marks)

Q27.

A set of three planes is given by the system of equations

$$x + 3y - z = 10$$

$$2x + ky + z = -4$$

$$3x + 5y + (k-2)z = k+4$$

where k is a constant.

- (a) Show that $\begin{vmatrix} 1 & 3 & -1 \\ 2 & k & 1 \\ 3 & 5 & k-2 \end{vmatrix} = k^2 - 5k + 6$. (2)
- (b) In each of the following cases, determine the **number** of solutions of the given system of equations.
 - (i) $k = 1$.
 - (ii) $k = 2$.
 - (iii) $k = 3$.(7)
- (c) Give a geometrical interpretation of the significance of each of the three results in

part (b) in relation to the three planes.

(3)

(Total 12 marks)

Q28.

(a) Evaluate the determinant of the matrix $\mathbf{M} = \begin{bmatrix} 1 & 3 & -1 \\ 2 & 3 & 1 \\ 4 & 0 & -5 \end{bmatrix}$.

(2)

(b) A three-dimensional shape S , with volume 12 cm^3 , is transformed by a transformation having matrix $\mathbf{X}$. Find the volume of the image of S in the case when:

(i) $\mathbf{X} = \mathbf{M}$;

(2)

(ii) $\mathbf{X} = \mathbf{M} \mathbf{N}^2$, where $\mathbf{N}$ is a 3×3 matrix and $\det \mathbf{N} = \frac{1}{3}$.

(2)

(Total 6 marks)

Q29.

(a) Show that $(a + b + c)$ is a factor of

$$\begin{vmatrix} a & b & c \\ b+c & c+a & a+b \\ b-c & c-a & a-b \end{vmatrix}$$

Express this determinant as the product of $(a + b + c)$ and a quadratic factor.

(5)

(b) Hence, or otherwise, show that there is a single real value of a for which the system of equations

$$ax + 3y + z = -5$$

$$4x + (1 + a)y + (a + 3)z = 9$$

$$2x + (1 - a)y + (a - 3)z = 15$$

does not have a unique solution, and find this value of a .

(5)

(Total 10 marks)